as to whether it is the average of first series or second series or third series or of any other series of 5 observations whose sum is 45. Thus we see that the measures of central tendency are inadequate to give us a complete idea of the distribution. So they must be supported and supplemented by some other measures. One such measure is Dispersion, which provides the nature of spreadness of the data.

Literal meaning of dispersion is "scatteredness" we study dispersion to have an idea about the homogeneity or heterogeneity of the distribution. In the above case we say that series (i) is more homogeneous (less dispersed) than the series (ii) or (iii) or we say that series (iii) is more heterogeneous (more scattered) than the series (i) or (ii).

Various measures of dispersion can be classified into two broad categories.

(a) The measures which express the spread of observations in terms of distance between the values of selected observations. These are also termed as distance measures.

Example: Range and interquartile range (or) quartile deviation.

(b) The measures which express the spread of observations in terms of the average of deviations of observation from some central value.

Example: Mean deviation and Standard deviation.

8.2.1 Quartile Deviation

Quartile Deviation is defined as

$QD = \frac{1}{2}(Q_3 - Q_1)$. It may also be called as semi-inter quartile. Where Q_1 and Q_3 are the first and third quartiles of the distribution respectively and $Q_3 - Q_1$ is called as inter quartile range.

(i) Relative measures for QD

Quartile deviation is an absolute measure of dispersion. The relative measure corresponding to this measure, called the coefficient of quartile deviation is calculated as follows:

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Coefficient of quartile deviation can be used to compare the degree of variation in different distributions.

(ii) Computation of Quartile Deviation

The process of computing quartile deviation is very simple since we just have to compute the values of the upper and lower quartiles that is Q_3 and Q_1 respectively.

Example 8.16

Calculate the value of quartile deviation and its coefficient from the following data

Roll No.	1	2	3	4	5	6	7
Marks	20	28	40	12	30	15	50

Solution:

Marks are arranged in ascending order

12 15 20 28 30 40 50

n = number of observations =7

Q_1= Size of $\left(\frac{(n+1)}{4}\right)^{th}$ value = Size of $\left(\frac{7+1}{4}\right)^{th}$ value = Size of 2nd value = 15

Hence $Q_1 = 15$

Q_3 = Size of $\left(\frac{3(n+1)}{4}\right)^{th}$ value = Size of $\left(\frac{3\times 8}{4}\right)^{th}$ value = Size of 6th value = 40

Hence $Q_3 = 40$

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